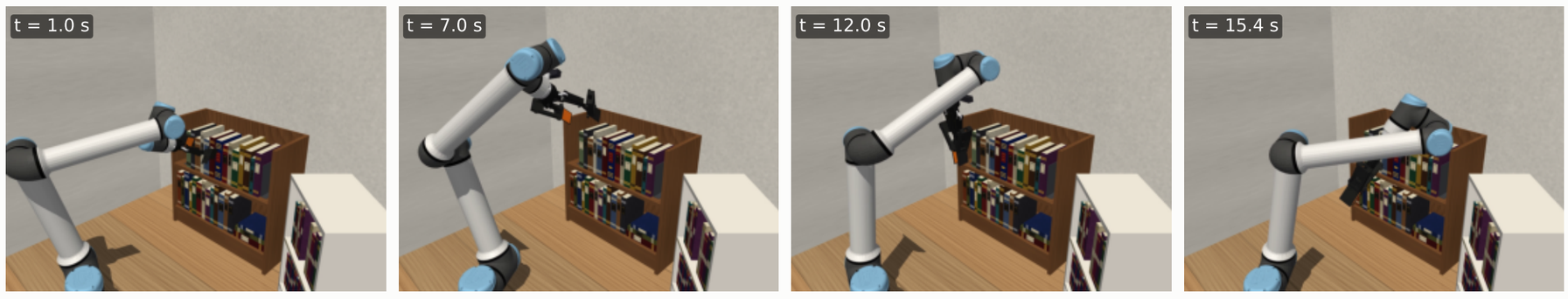
Why each row of the results table scores what it does: frames of re-runs of evaluation test cases (views: whole cell, 3/4 close-up, top of the shelf; the gripper enters from the bottom in the top view)

π0.5-DROID zero-shot · nominal

wanders above the shelf with the jaw open, then sweeps into it: wrist force > 150 N = UR protective stop

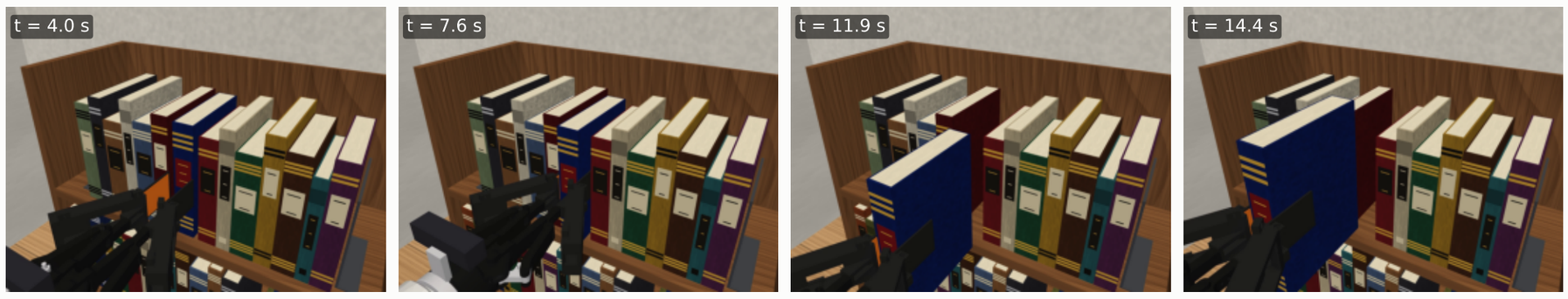
→ 0/20 nominal, 0/20 randomized (19 of 20 stop like this)



π0.5 fine-tuned · nominal (its training scene)

blades into the 10 mm gaps, squeeze, pull out, lift, hold 1 s

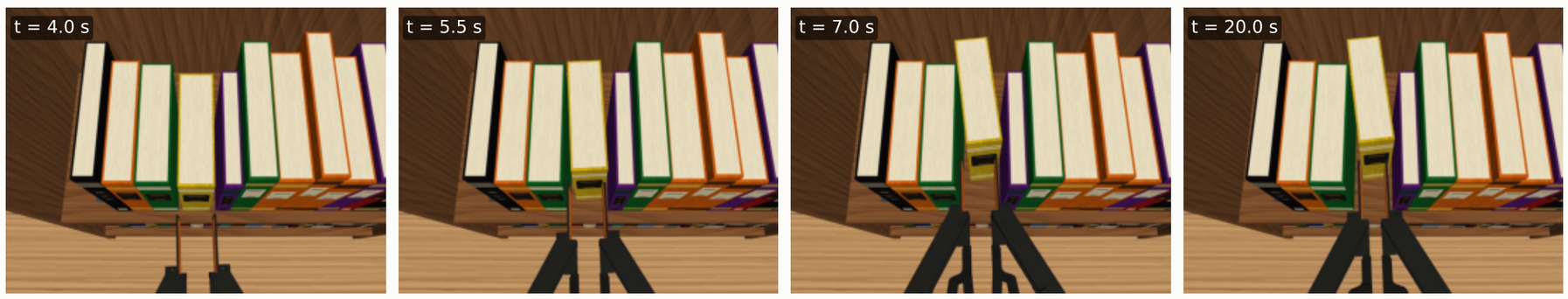
→ 50/50, neighbours untouched (median 0.0 mm)



π0.5 fine-tuned · randomized, 45 mm book

opens only 2 mm wider than the book (expert: 6 mm); the blade tips catch the spine and push it 56 mm deeper; the pads never touch it

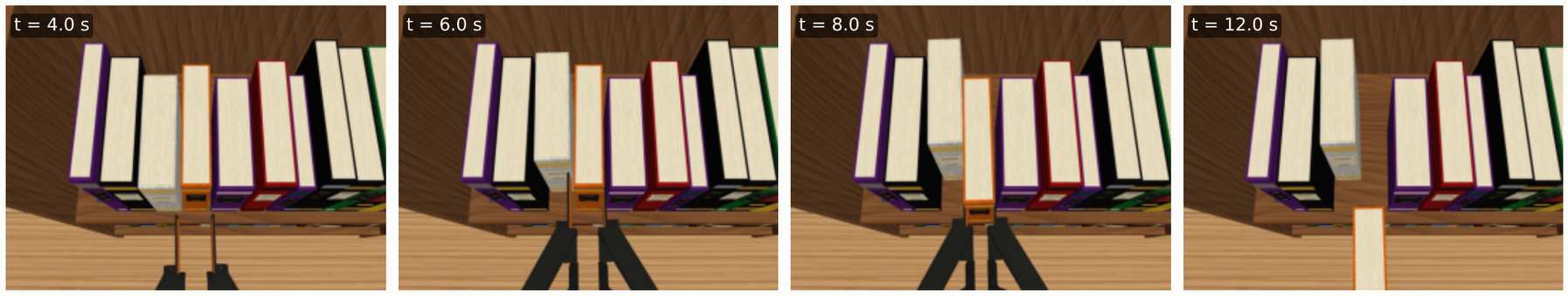
→ 36/50: 7 of the 14 failures are push-ins like this



π0.5 fine-tuned · narrow gaps (7 mm)

opens to its 40 mm-book setting around a 36 mm book: the blades need 53 mm, the slot is 51 mm, so a neighbour is shoved 56 mm (still a success)

→ 22/30, but a neighbour moved > 5 mm in 60 % of episodes



π0.5 fine-tuned · heavy + slippery (2.35 kg, μ 0.5)

pulls it out, then the book rotates (39°) and slips out of the fixed 67 N grip during the lift

→ 13/30: 12 of the 17 failures slip in the lift



scripted expert · same test case

perfect insertion (0.0 mm), same fixed grip: the book slips too

→ 17/30: even perfect perception loses 13 of 30 with this fixed grip

